🏦 Institutional NAC Pre-Vetting & Risk Mitigation Framework

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Project: PoC 21 - Portfolio Optimization & Risk Mitigation
Tech Stack: Python | Streamlit | LangChain (RAG) | ChromaDB | Groq (Llama 3.3-70B)

📋 Project Vision

In institutional banking, the New Activity (NAC) process is often a strategic bottleneck. This Proof of Concept (PoC) demonstrates how Retrieval-Augmented Generation (RAG) can shift governance from a reactive "No" to a proactive "Pre-Audit."

The tool acts as a Digital First-Responder, allowing Managing Directors to vet partnership proposals against complex internal policies in seconds, identifying regulatory "red flags" and financial misalignments before they reach the committee.

🏛️ Executive Summary

Strategic Intent: To transform the 'New Activity' (NAC) process from a reactive bottleneck into a proactive strategic filter.

The Problem

Institutional deal teams often spend weeks preparing partnership proposals, only to face "late-stage rejection" due to governance or risk misalignments. This lag increases operational friction and slows the institutional deal pipeline.

The Solution

A Retrieval-Augmented Generation (RAG) framework that acts as a "Digital First-Responder." By cross-referencing multi-million-dollar deal proposals against internal bank policies in real-time, the tool identifies regulatory 'red flags' and provides an instant go/no-go risk assessment.

Key Value Drivers

- **Instant Visibility:** A dynamic Risk Heatmap provides immediate clarity on Financial, Regulatory, and Operational exposure.
- **Explainable AI (XAI):** Every finding is backed by an indisputable audit trail, quoting exact policy snippets from the internal knowledge base.
- **Operational Efficiency:** Drastically reduces committee review cycles by ensuring only "governance-ready" proposals move forward.

✨ Key Features

- **Dual-Engine RAG:** Combines deep narrative audit analysis with a dedicated quantitative scoring engine.
- **Risk Exposure Heatmap:** A color-coded visual dashboard (Yellow-Orange-Red) for instant risk quantification.
- **Indisputable Audit Trail:** Every finding includes Source Snippets—direct quotes from internal governance PDFs to ensure 100% transparency.
- **Executive Export:** Generates professional, board-ready PDF audit reports for formal submission.
- **Scalable Knowledge Base:** Modular design allows for instant policy updates via simple PDF ingestion without model retraining.

🛠️ Technical Architecture

- **Orchestration:** LangChain (RAG Architecture)
- **LLM:** Llama 3.3-70B (via Groq) for high-speed institutional reasoning.
- **Vector Database:** ChromaDB (Local persistent storage).
- **Embeddings:** HuggingFace all-MiniLM-L6-v2 (100% local/private processing).
- **Frontend:** Streamlit (Optimized for Managing Director-level UX).

🚀 Scalability Roadmap: From PoC to Production

- **Multi-Agent Review:** Deploy specialized "AI Agents" for specific risk domains (e.g., a "Legal Agent" and a "Financial Audit Agent") to provide deeper cross-functional analysis.
- **Enterprise Auth:** Integration with Okta/Azure AD for secure, role-based access to sensitive governance documents.
- **Audit Trail Logging:** Connecting the tool to an immutable database (e.g., SQL or Blockchain) to log every pre-audit for future regulatory reviews.
- **Feedback Loop:** Allowing Risk Officers to "thumbs up/down" AI findings to fine-tune the RAG retrieval precision over time.

🛠️ Installation & Usage

1. **Clone & Setup:**

```
pip install -r requirements.txt
pip install fpdf2 seaborn matplotlib pandas
```

2. **Launch Portal:**

```
streamlit run app.py
```

Use code with caution.

3. **Analyze:** Enter your Groq API key, ingest your policy library (/data/governance_docs), and upload a proposal to generate your first audit.

📊 Dashboard Strategy

The UI is optimized for Managing Director review, utilizing a 1:2 layout ratio that prioritizes a narrow, high-impact Risk Dashboard on the left and a detailed, high-readability Policy Narrative on the right.

📖 User Guide: NAC Pre-Vetting Portal

Role: Self-Service Strategic Alignment & Risk Identification

1. Getting Started

- **Authentication:** Enter your Groq API Key in the sidebar. This connects the secure "reasoning engine" to the portal.
- **Knowledge Sync:** If new governance policies have been released, click "Update/Ingest Governance Docs." The system will scan the internal library and refresh its "memory" (this takes about 30–60 seconds).

2. Running an Assessment

- **Upload:** Drag and drop your Partnership Proposal (PDF or TXT) into the upload zone.
- **Analyze:** Click "🚀 Run Risk Assessment."
- **What happens next:** The AI performs two tasks. First, it searches the policy library for relevant rules. Second, it generates the detailed audit and calculates the risk scores.

3. Reading the Dashboard (Left Column)

- **The Risk Bar:** Look at the heatmap for an immediate "Go/No-Go" signal.
    - 🟢 Green (0-3): Strong alignment. Likely to pass committee with minimal questions.
    - 🟡 Yellow (4-6): Potential friction. Requires the deal team to provide additional justification or waivers.
    - 🔴 Red (7-10): Policy breach. The proposal likely violates a "Hard Rule" (e.g., Data Sovereignty or IRR floors) and should be revised before submission.
- **Quick-Check:** Use the icon list for a 5-second summary of the three core risk categories.

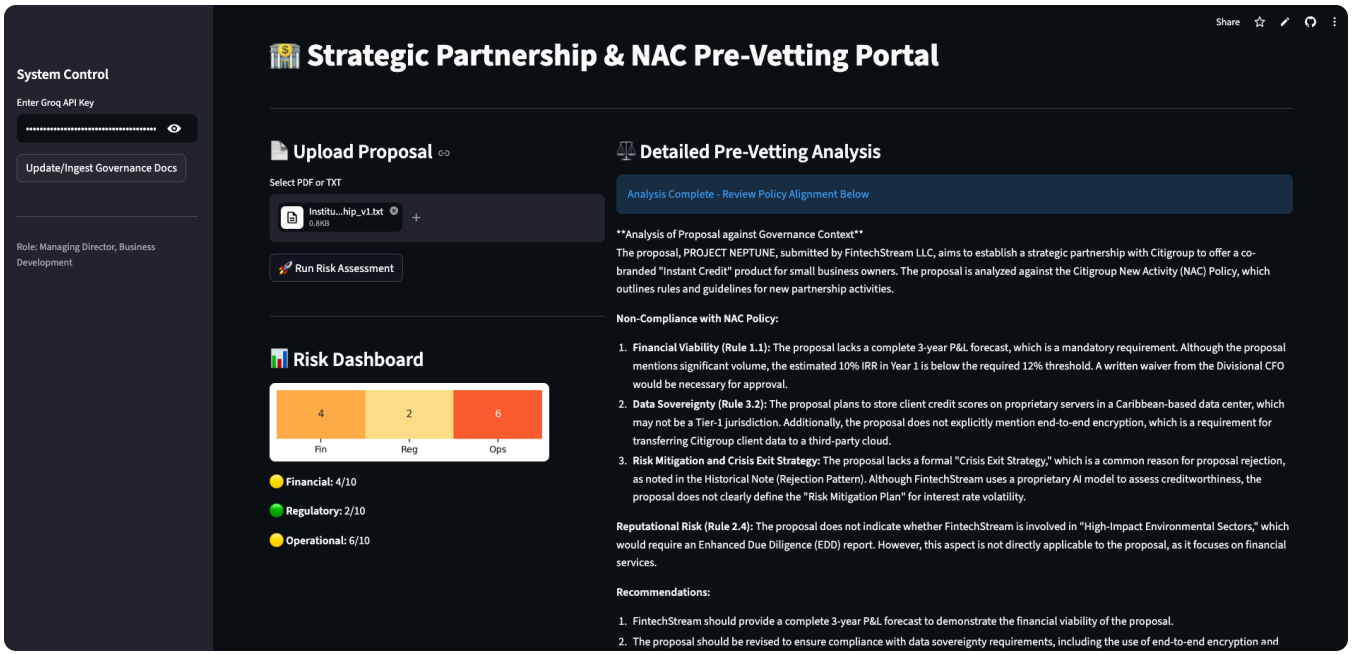
4. Reviewing the Deep-Dive (Right Column)

- **Executive Summary:** Read the top section for the AI's final verdict and a high-level recommendation.
- **Policy Snippets:** This is the most critical section. Look for the "Source Snippets" where the AI quotes the exact bank rule it is using to flag your proposal. This ensures the feedback is objective, not subjective.
- **Actionable Feedback:** Use the bullet points at the bottom of the report to refine your proposal. These are the specific steps needed to turn a "Red" or "Yellow" score into a "Green."

5. Exporting for Formal Submission

- **Download PDF:** Once you are satisfied with the alignment, click "📥 Download Audit Report."
- **Best Practice:** Attach this PDF to your formal NAC submission. It demonstrates to the Risk Committee that you have performed pre-submission due diligence and have already addressed potential misalignments.

📺 Screenshots



Risk Assessment Report: Project Neptune

1. **Executive Summary** (Go/No-Go Recommendation)

Based on the analysis of the proposal against the Citigroup New Activity (NAC) policy, I recommend a **No-Go** decision for Project Neptune in its current form. The proposal lacks critical components, such as a comprehensive 3-year P&L forecast, a clear crisis exit strategy, and adherence to data sovereignty requirements.

2. **Policy Alignment**

The proposal partially aligns with the NAC policy, but significant gaps exist:

* **Rule 1.1: Financial Viability** - The estimated 10% IRR in Year 1 falls short of the required 12% threshold, and the lack of a comprehensive 3-year P&L forecast is a concern.
* **Rule 2.4: Reputational Risk (Third-Party)** - Although FintechStream is not explicitly mentioned as operating in a "High-Impact Environmental Sector," the partnership may still pose reputational risks if not properly assessed.
* **Rule 3.2: Data Sovereignty** - The proposal does not meet the requirement for end-to-end encryption and hosting within a Tier-1 jurisdiction, as the data center is located in the Caribbean.

3. **Identified Risks**

The following risks have been identified:

* **Insufficient financial viability**: The proposed IRR is below the required threshold, and the lack of a comprehensive 3-year P&L forecast increases uncertainty.
* **Reputational risk**: The partnership with FintechStream may pose reputational risks if not properly assessed and mitigated.
* **Data sovereignty and security risks**: The storage of client credit scores in a Caribbean-based data center